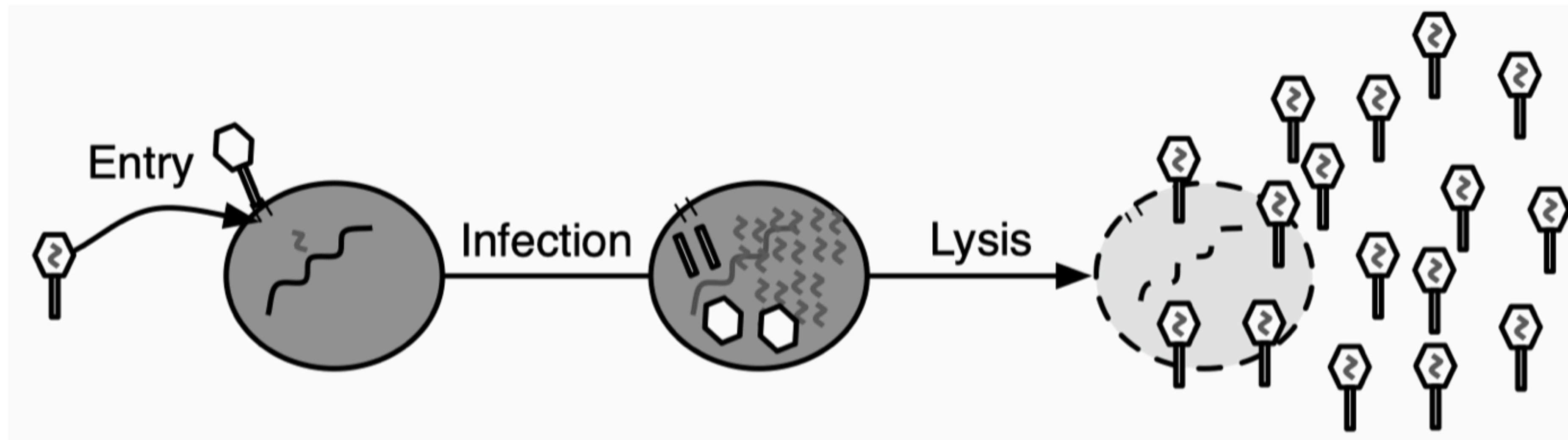


Eco-Evo dynamics

Evolution & Ecology

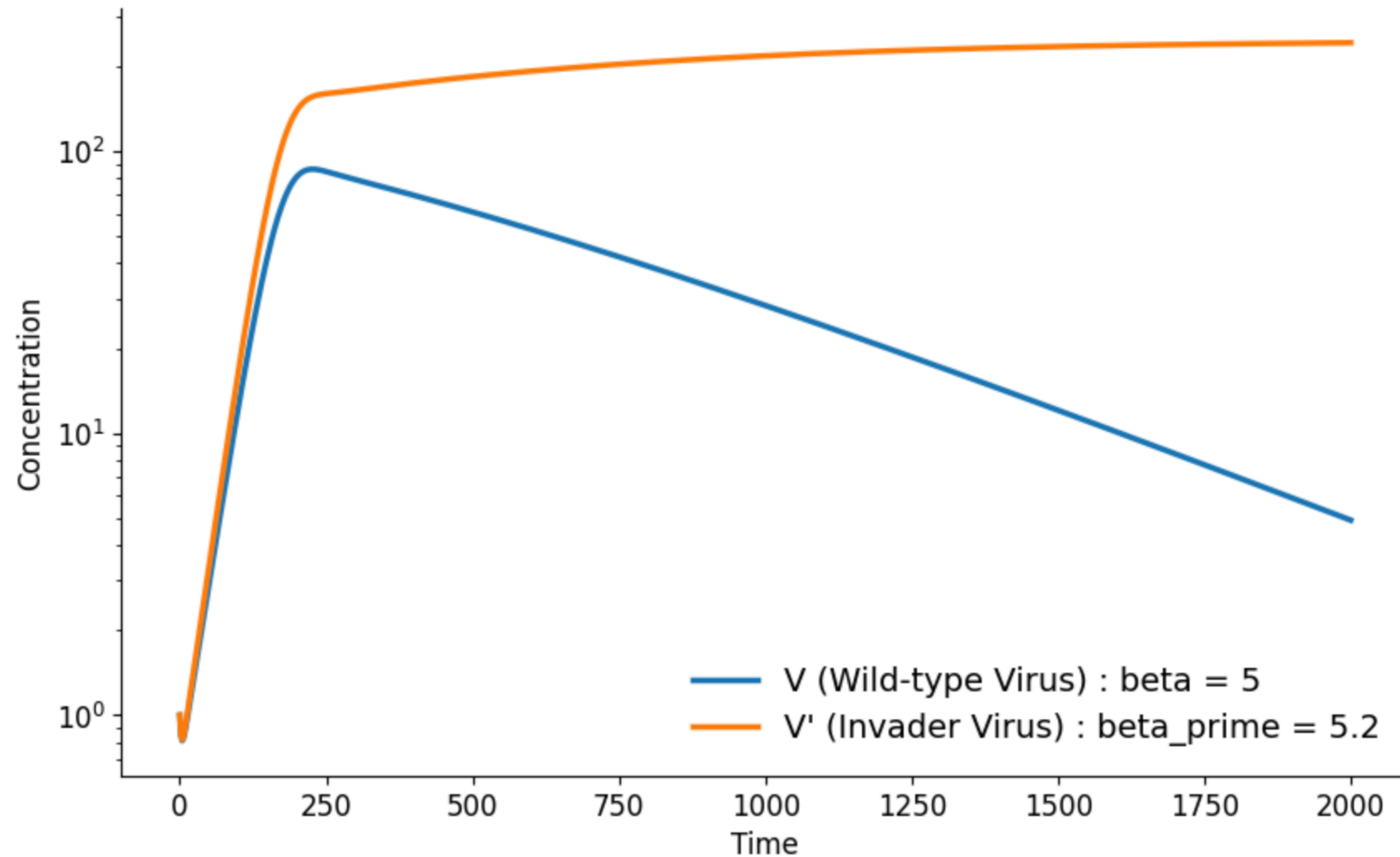
Jacopo Grilli, Ciudad de Guatemala, 1/12/2025

What is optimal?



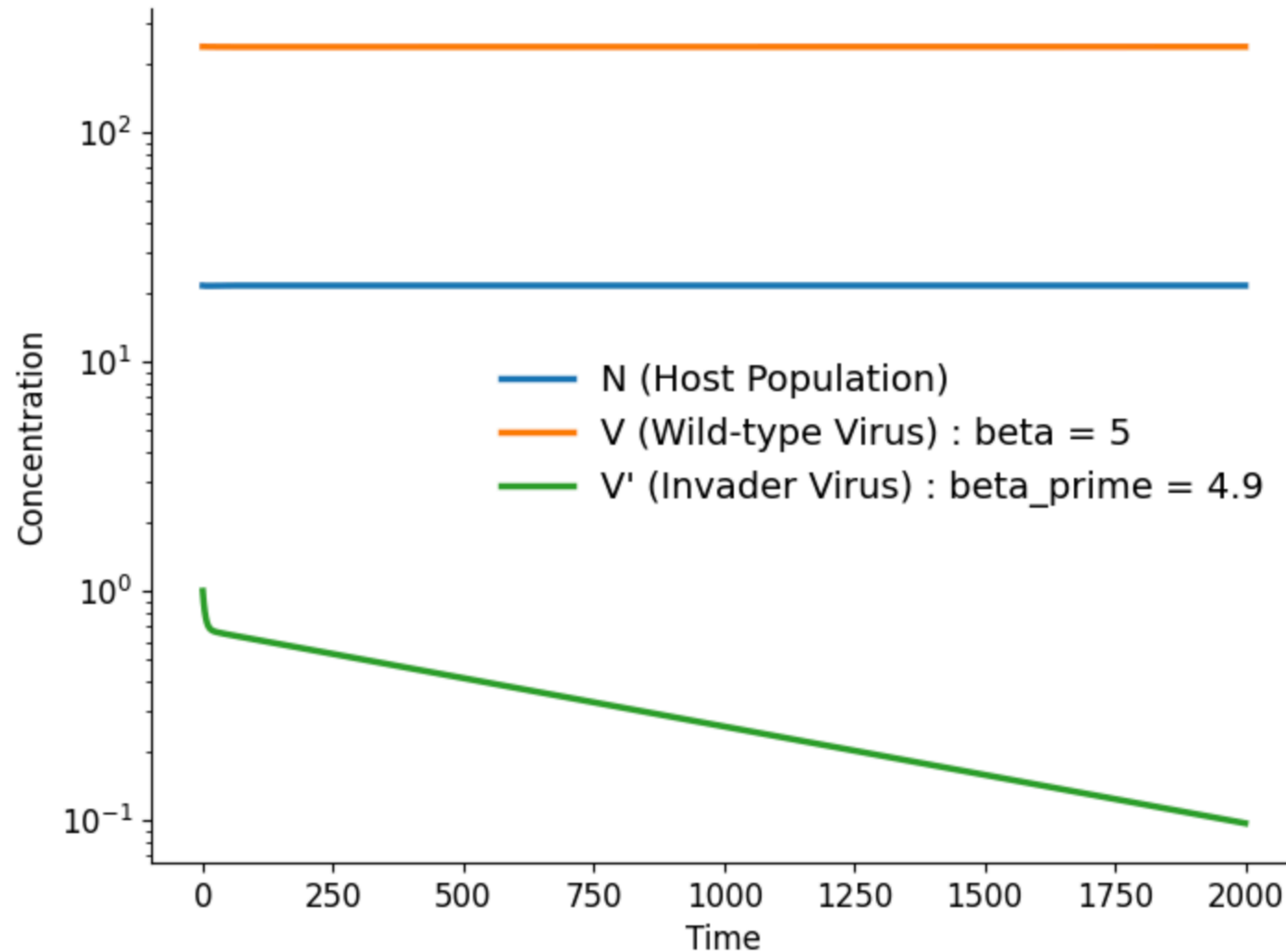
Multiple phages

Competition between phages

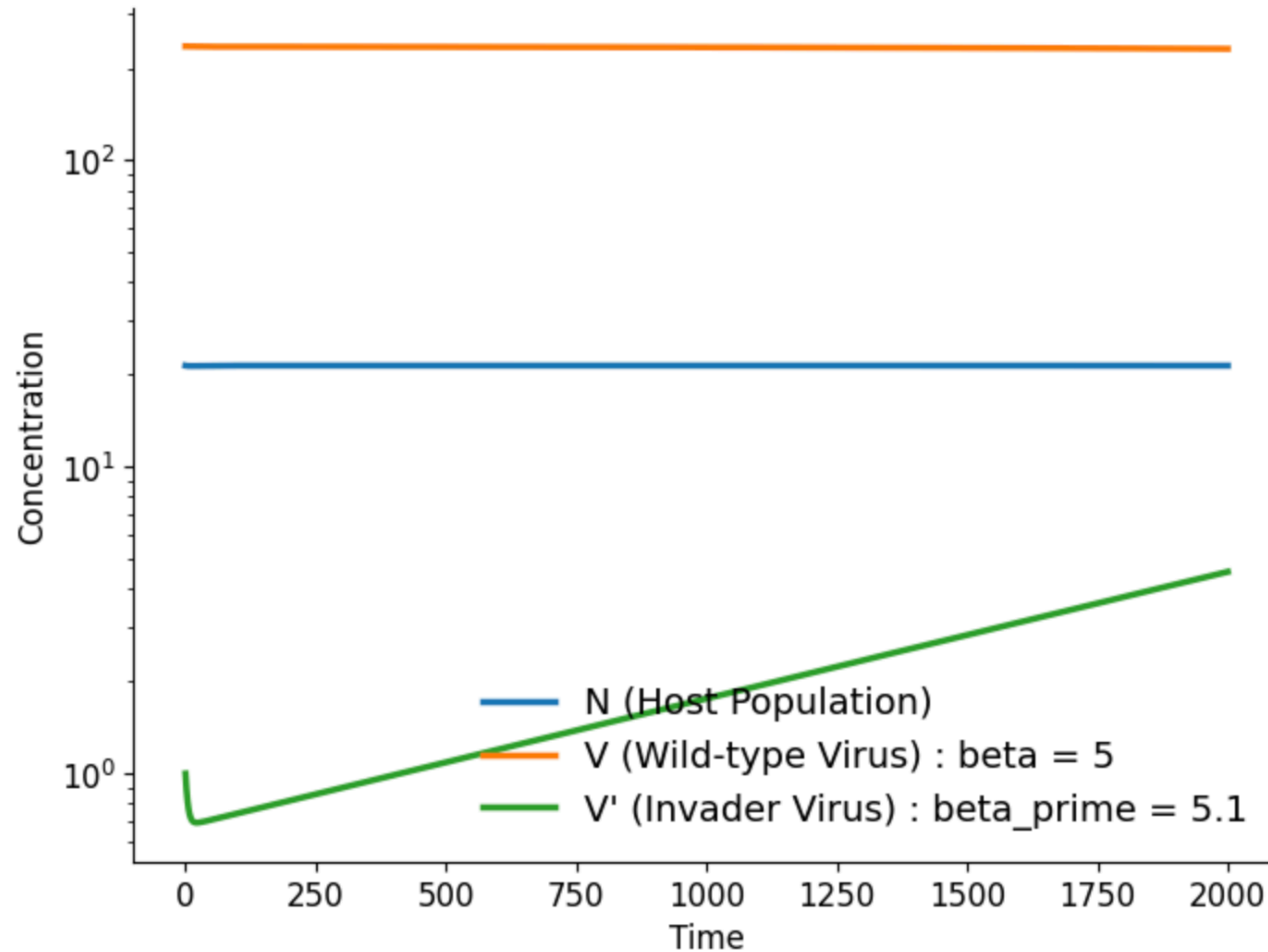


Invasion

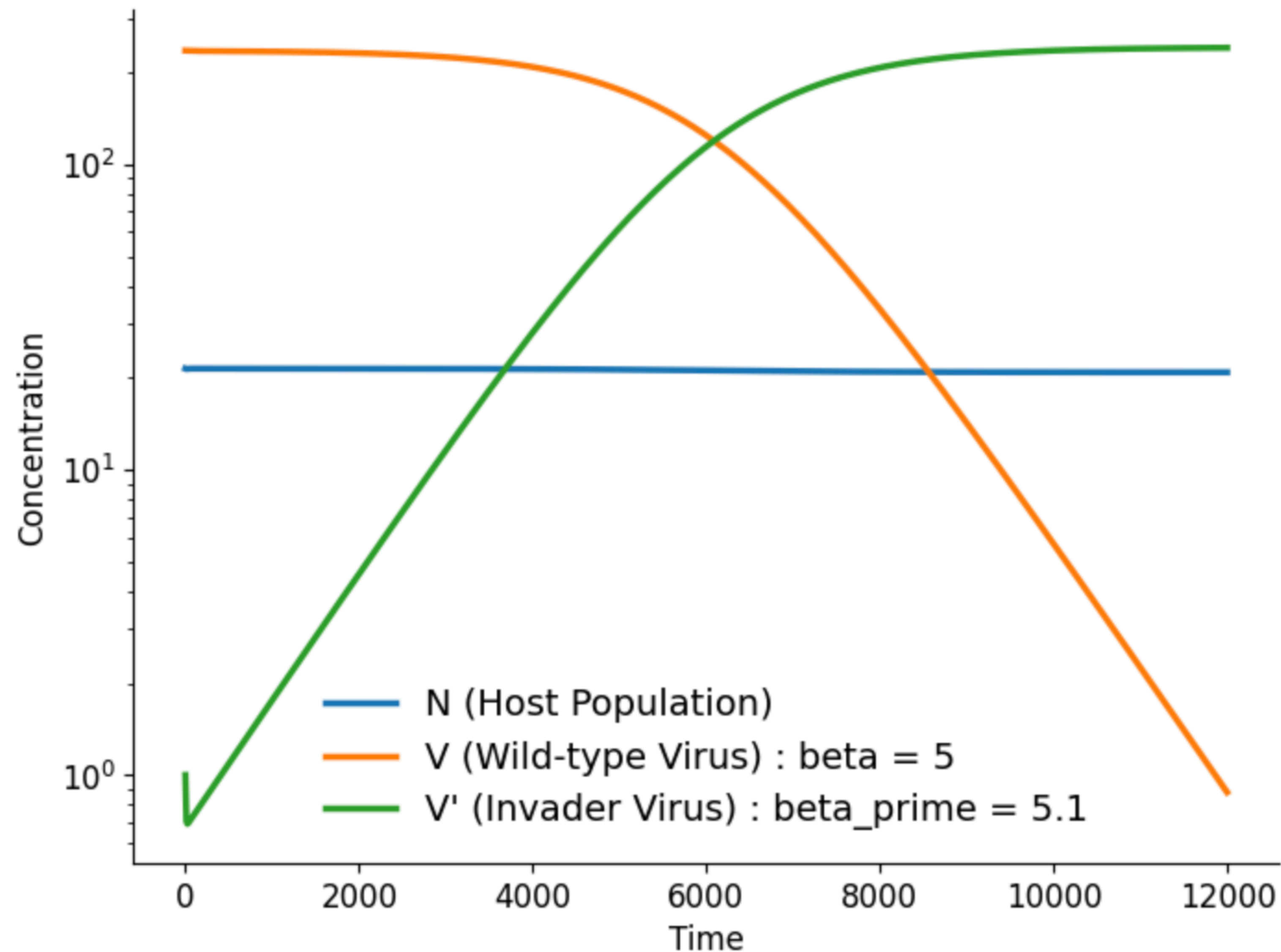
Failed invasion



Successful invasion

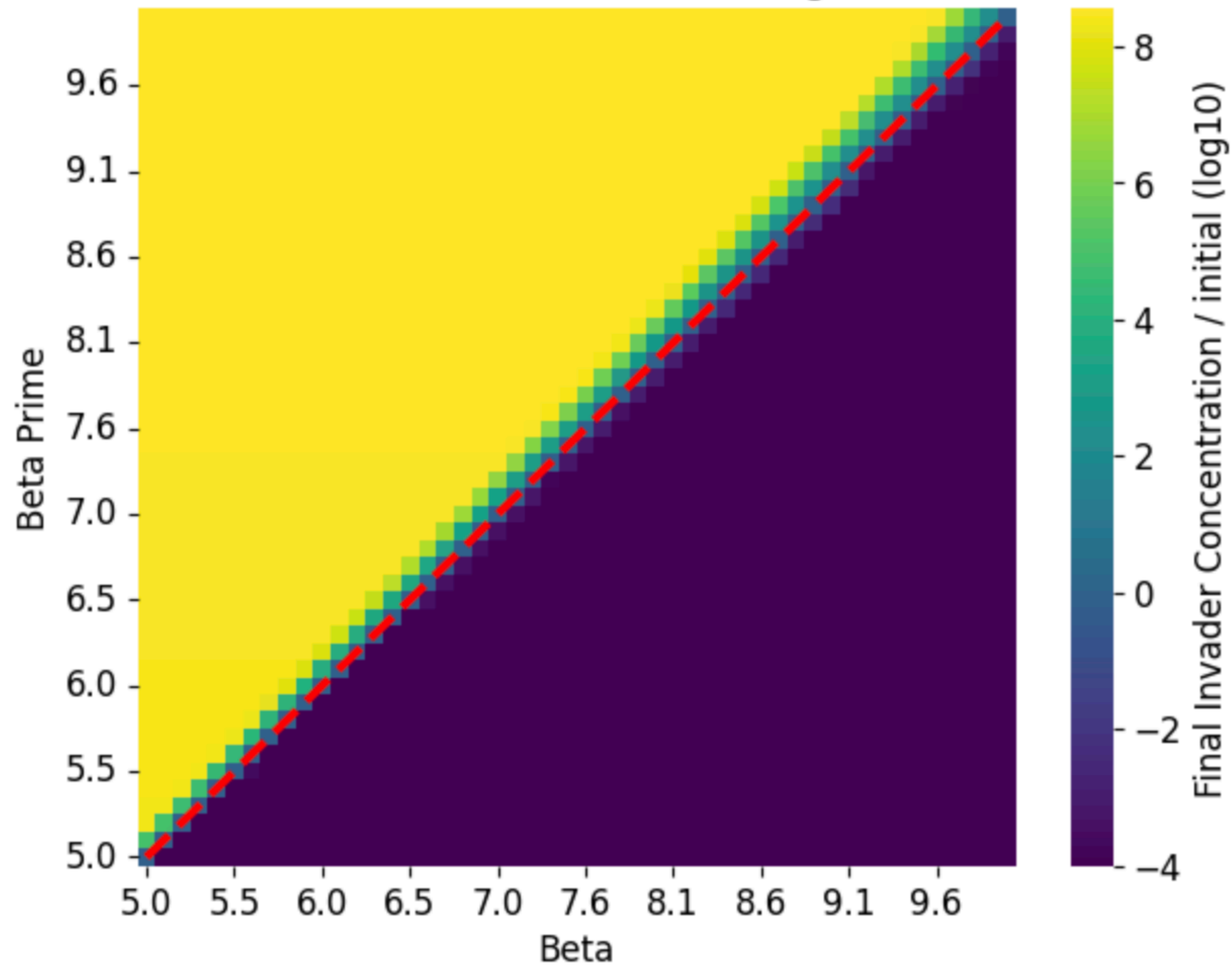


Successful invasion (longer time)

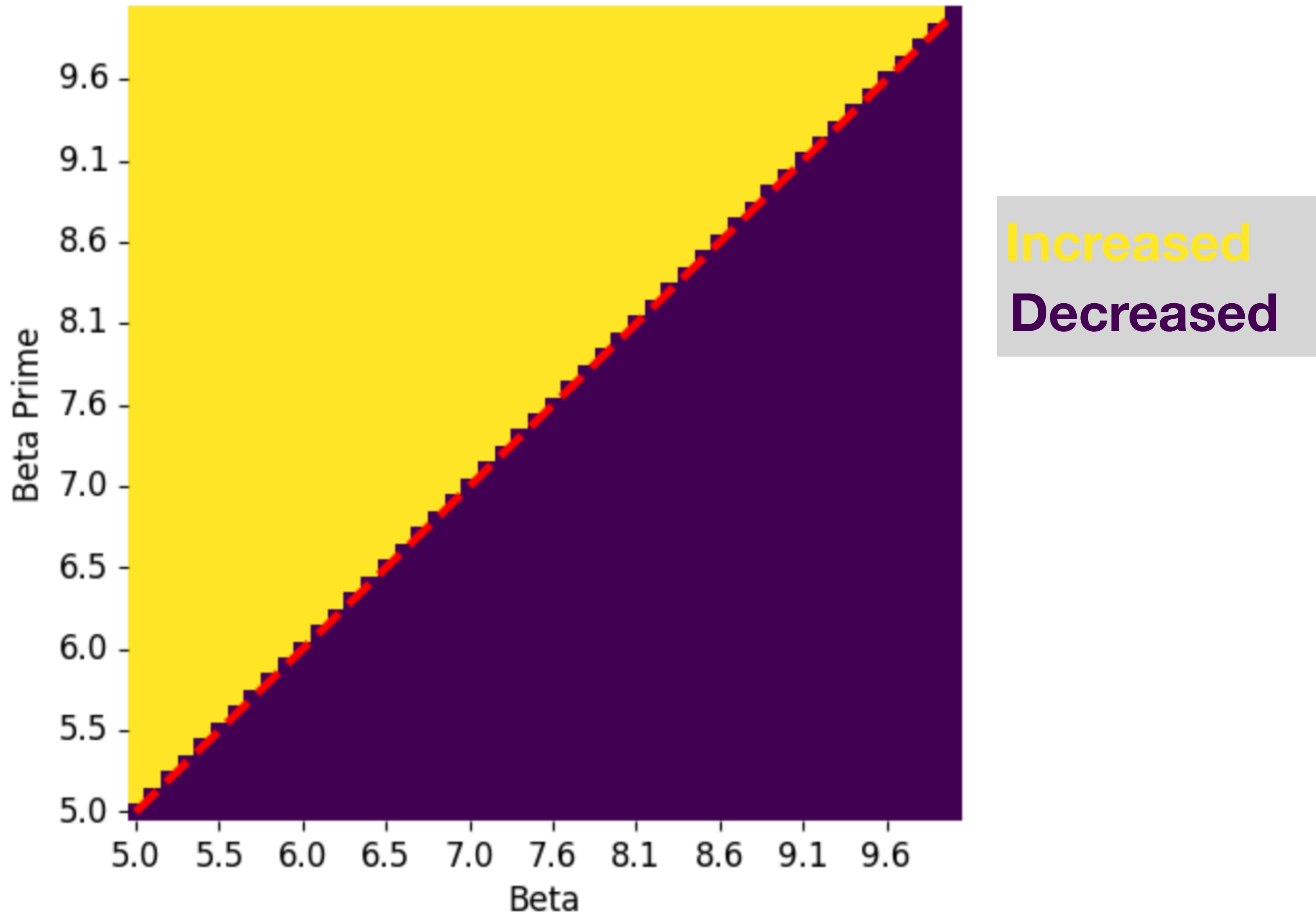


Invasions, systematically

Final invader concentration

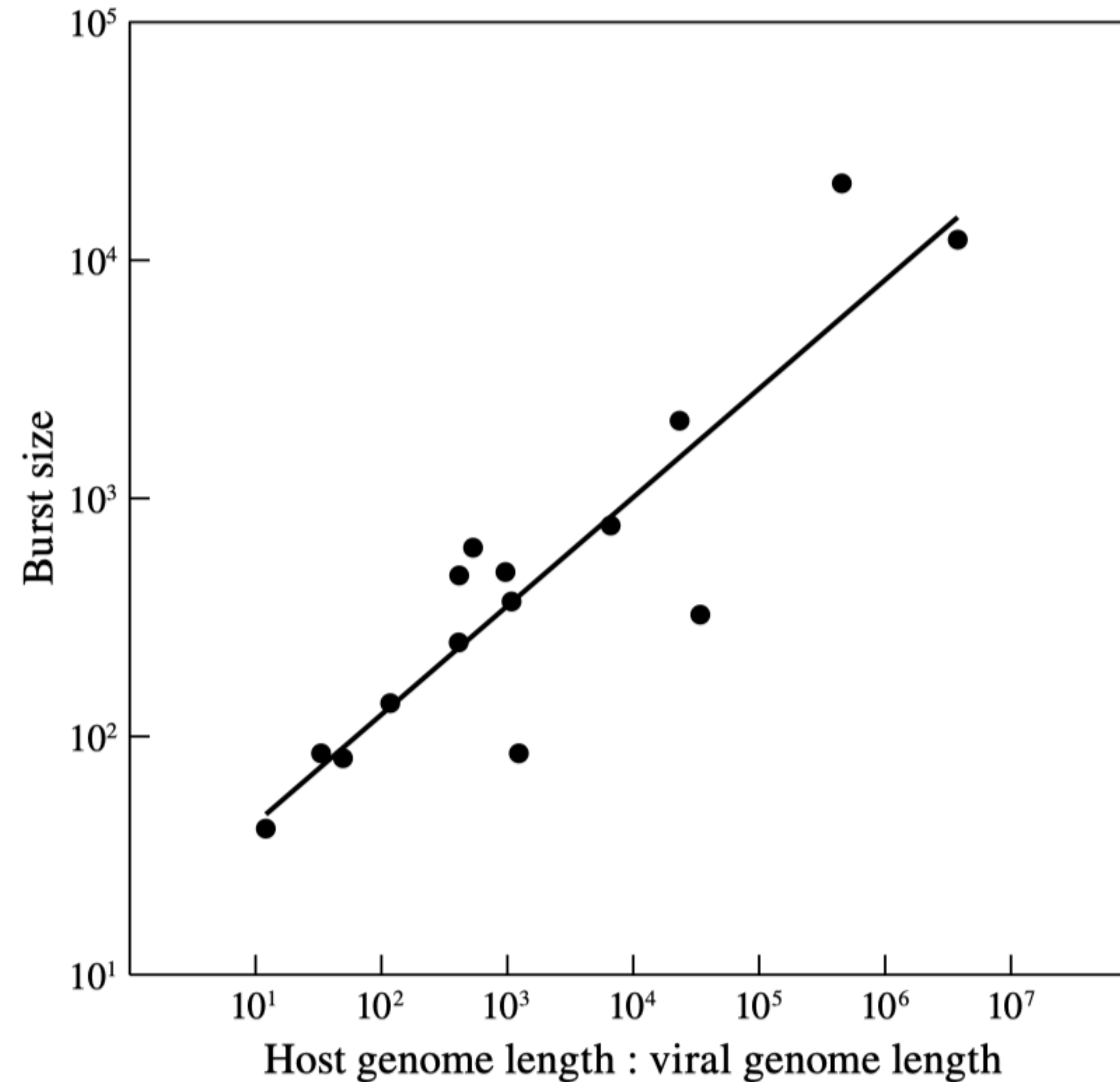
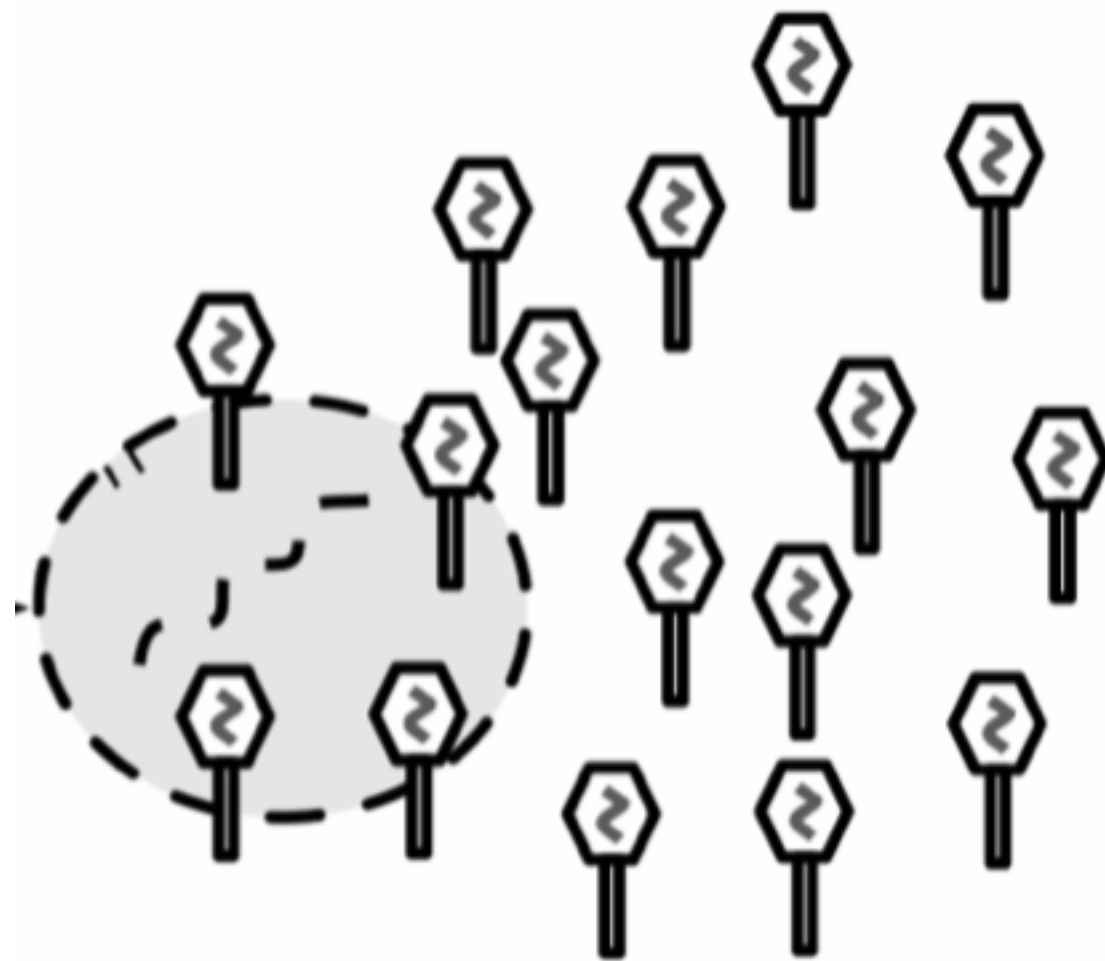


Final invader concentration. Did it grow?

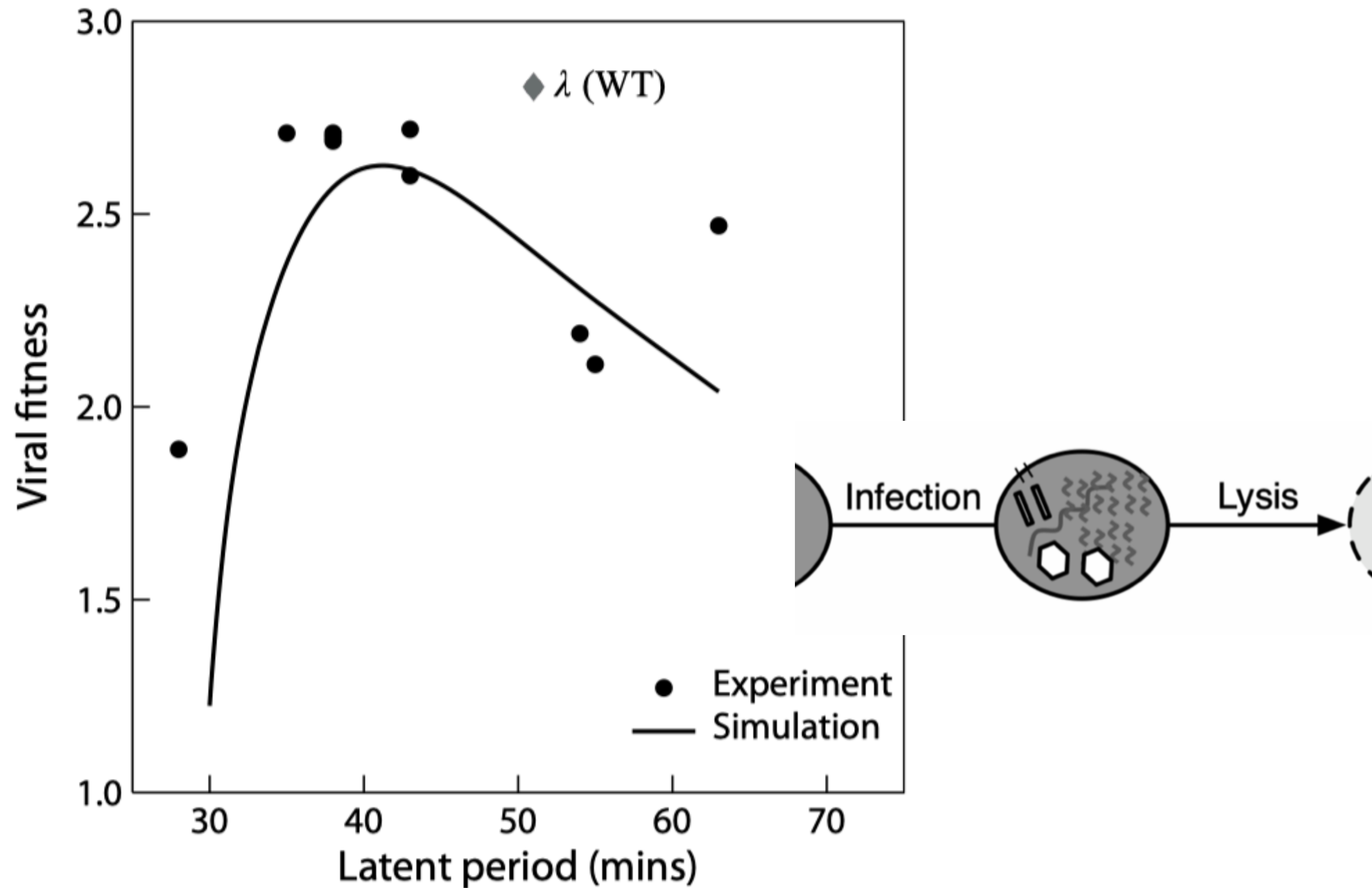


Optimal: maximize burst size
Is this the whole story?

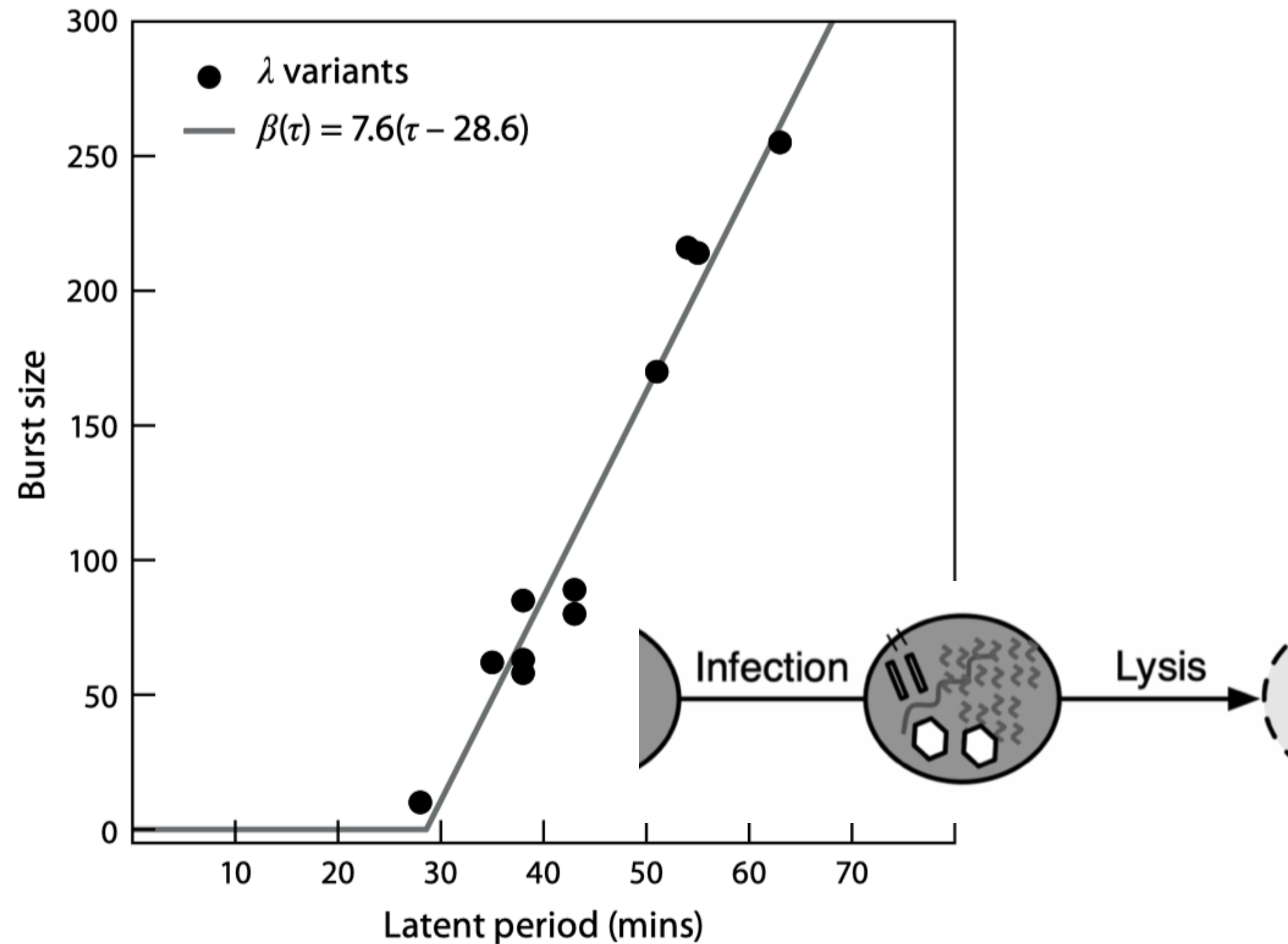
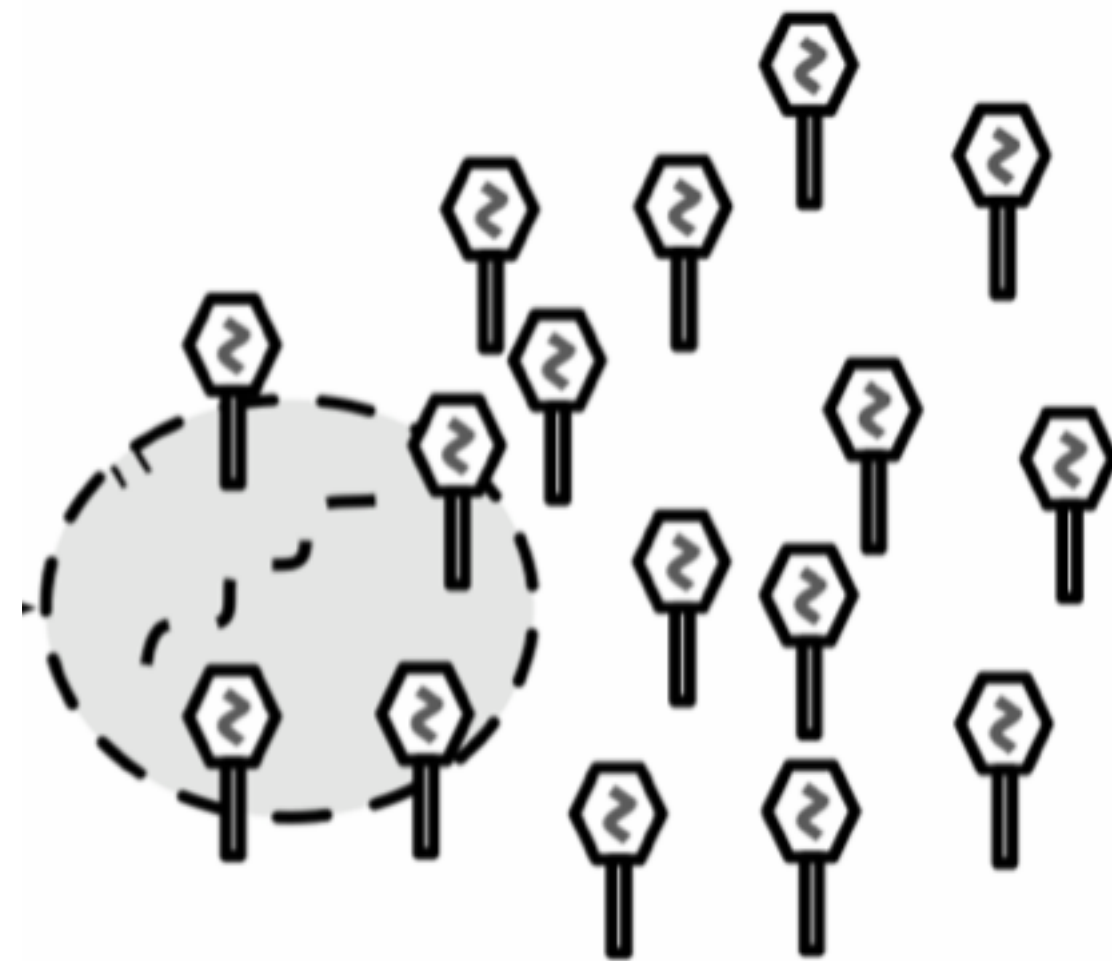
Variation of the burst size



Intermediate optimal latency time

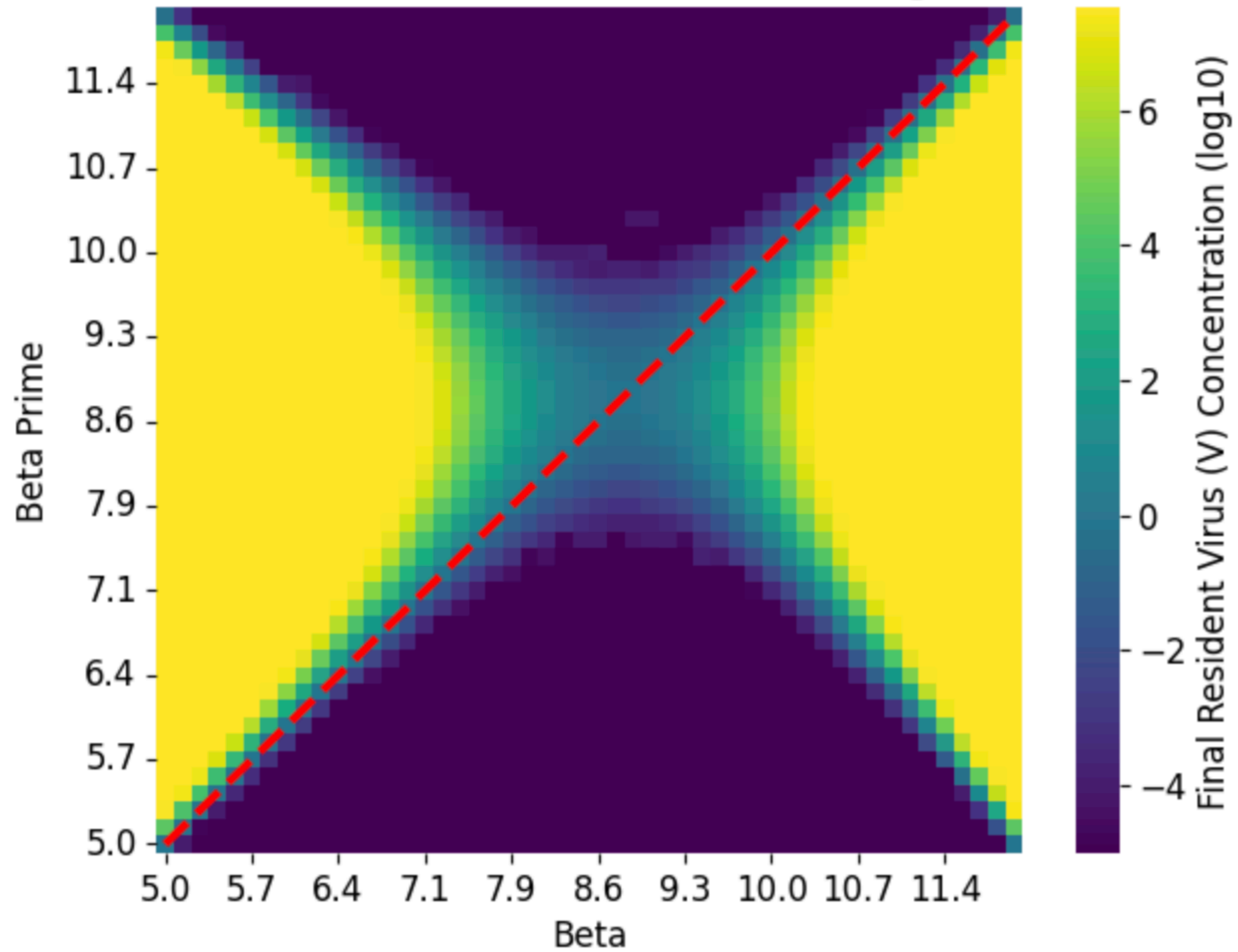


Latency time and burst are not independent

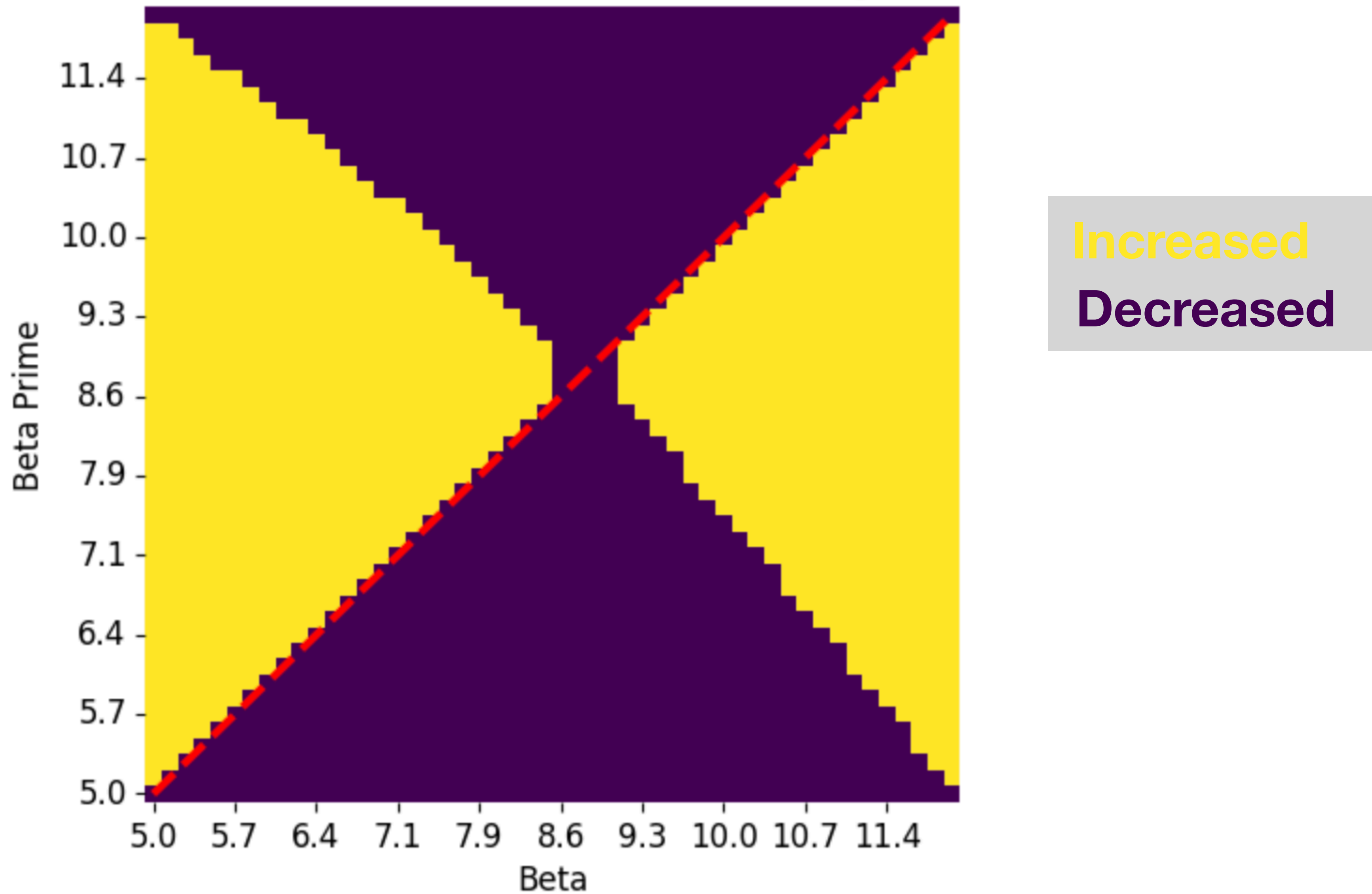


Tradeoff

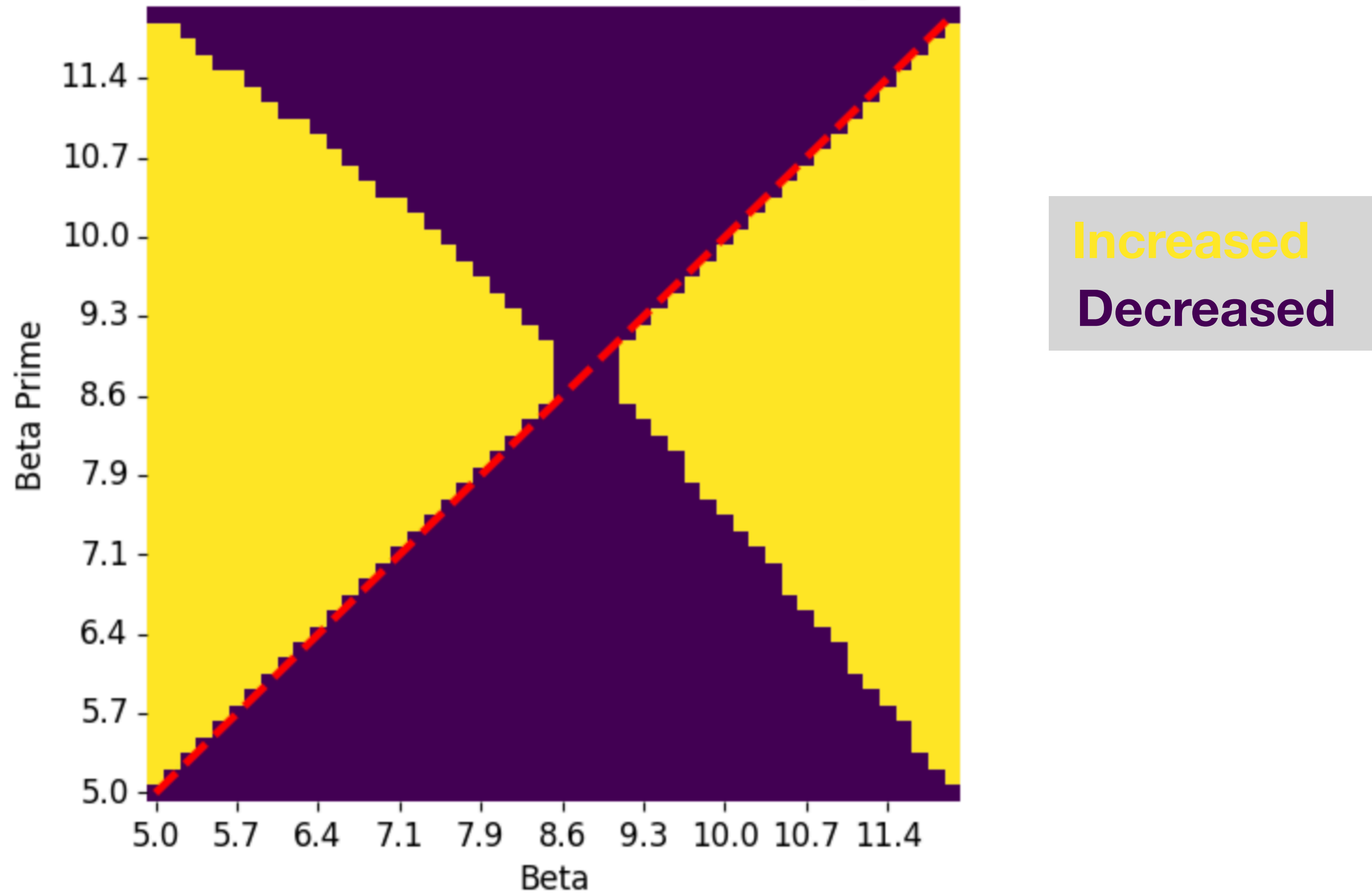
Final invader concentration



Final invader concentration. Did it grow?

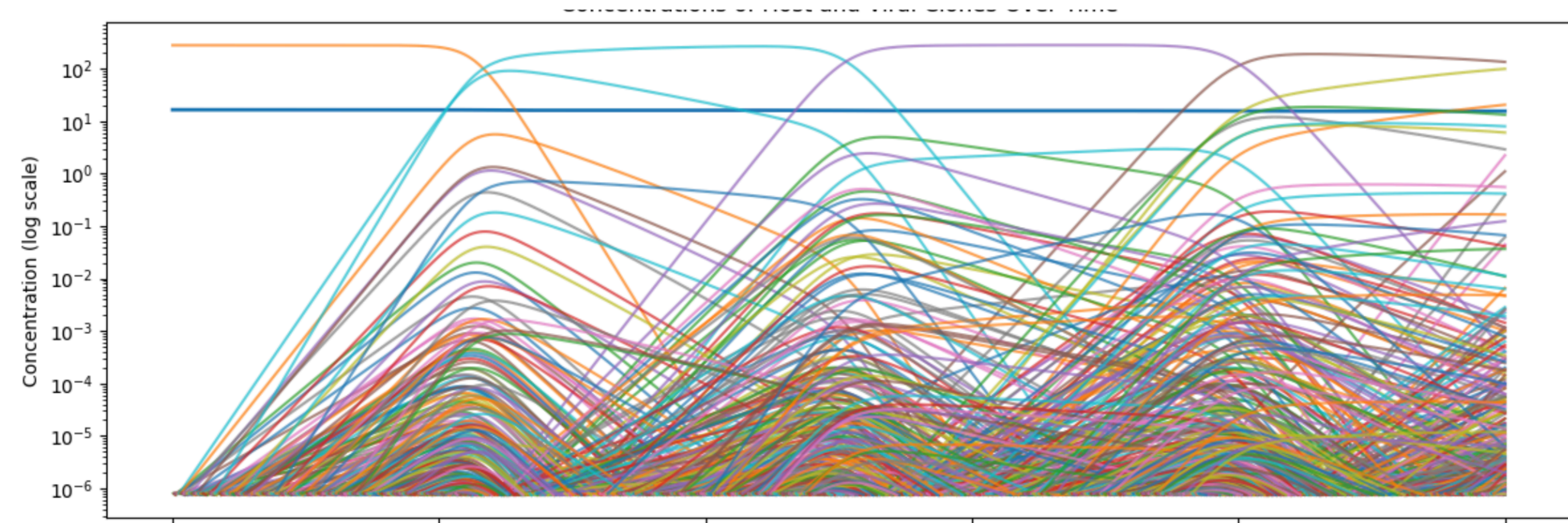


What do you expect over long time?

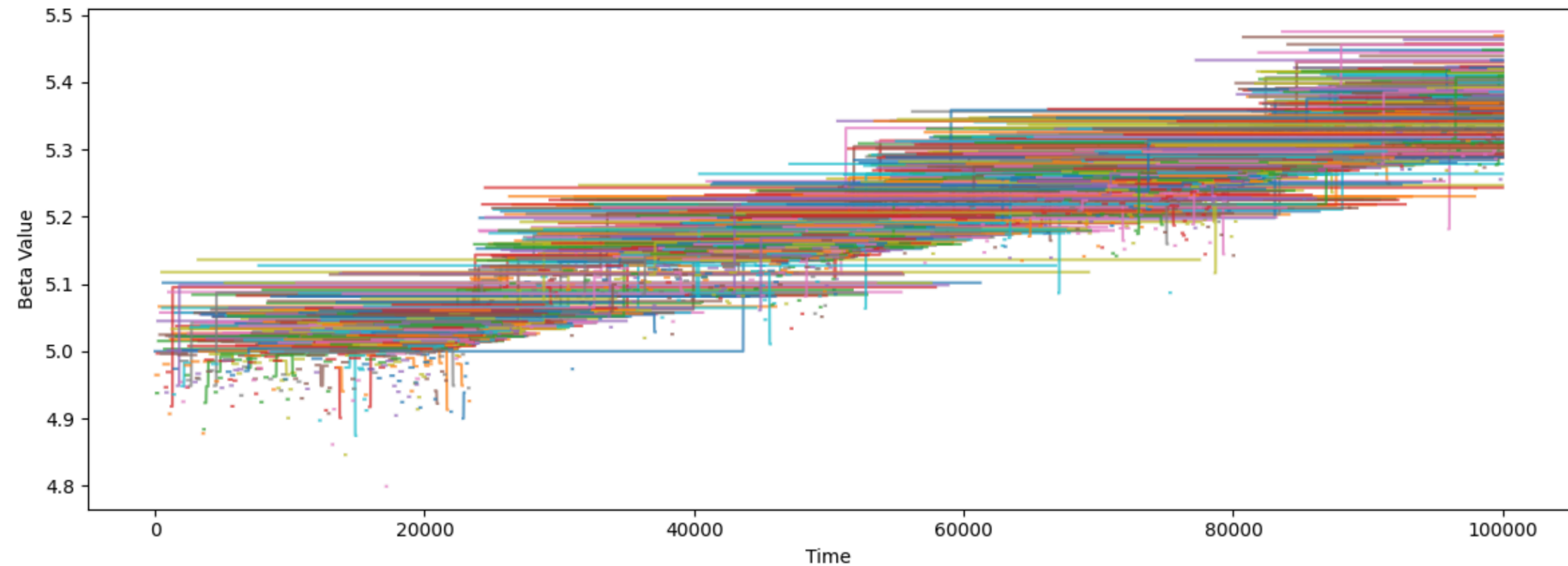


Invasions, systematically

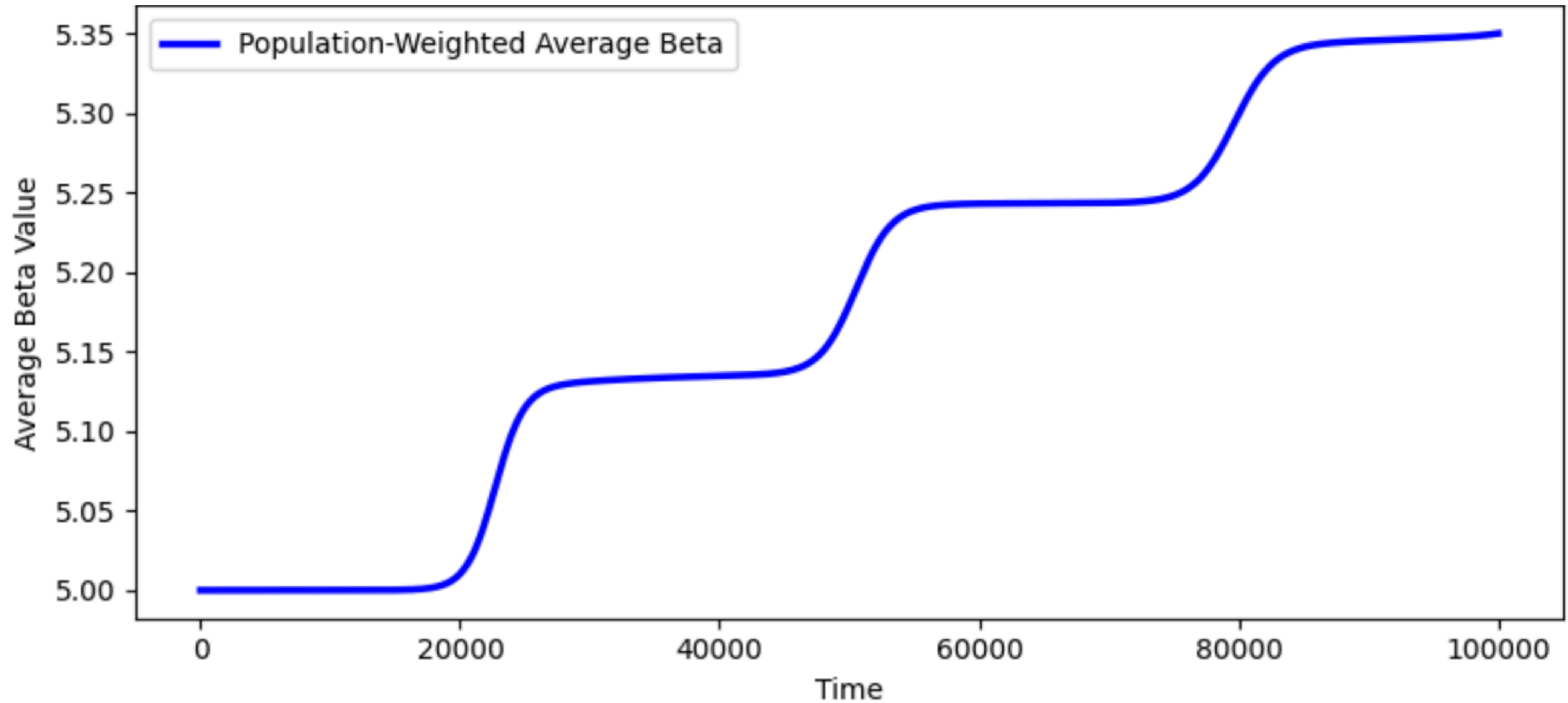
Evolutionary trajectories



Evolutionary trajectories



Evolutionary trajectories



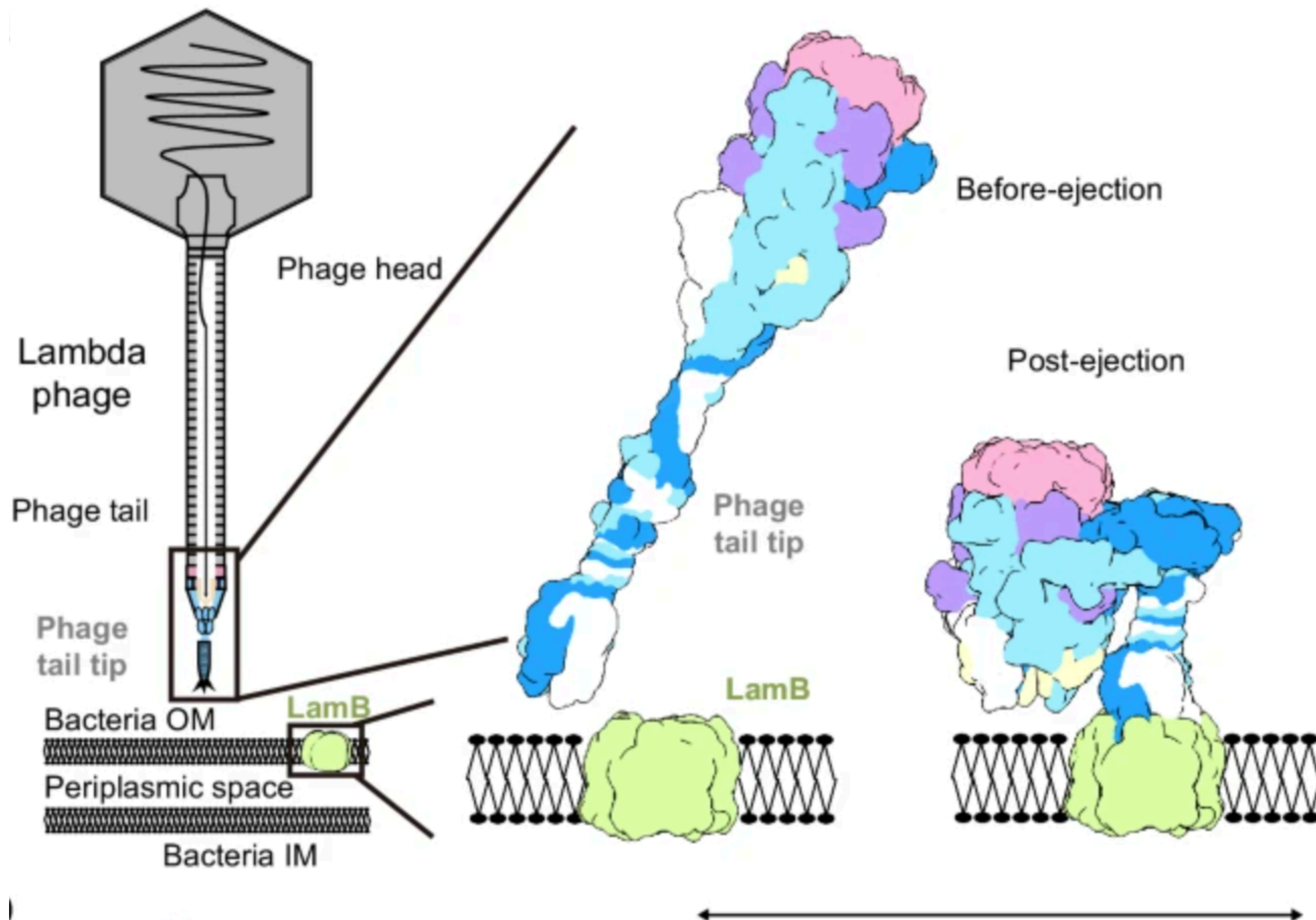
Adaptive dynamics

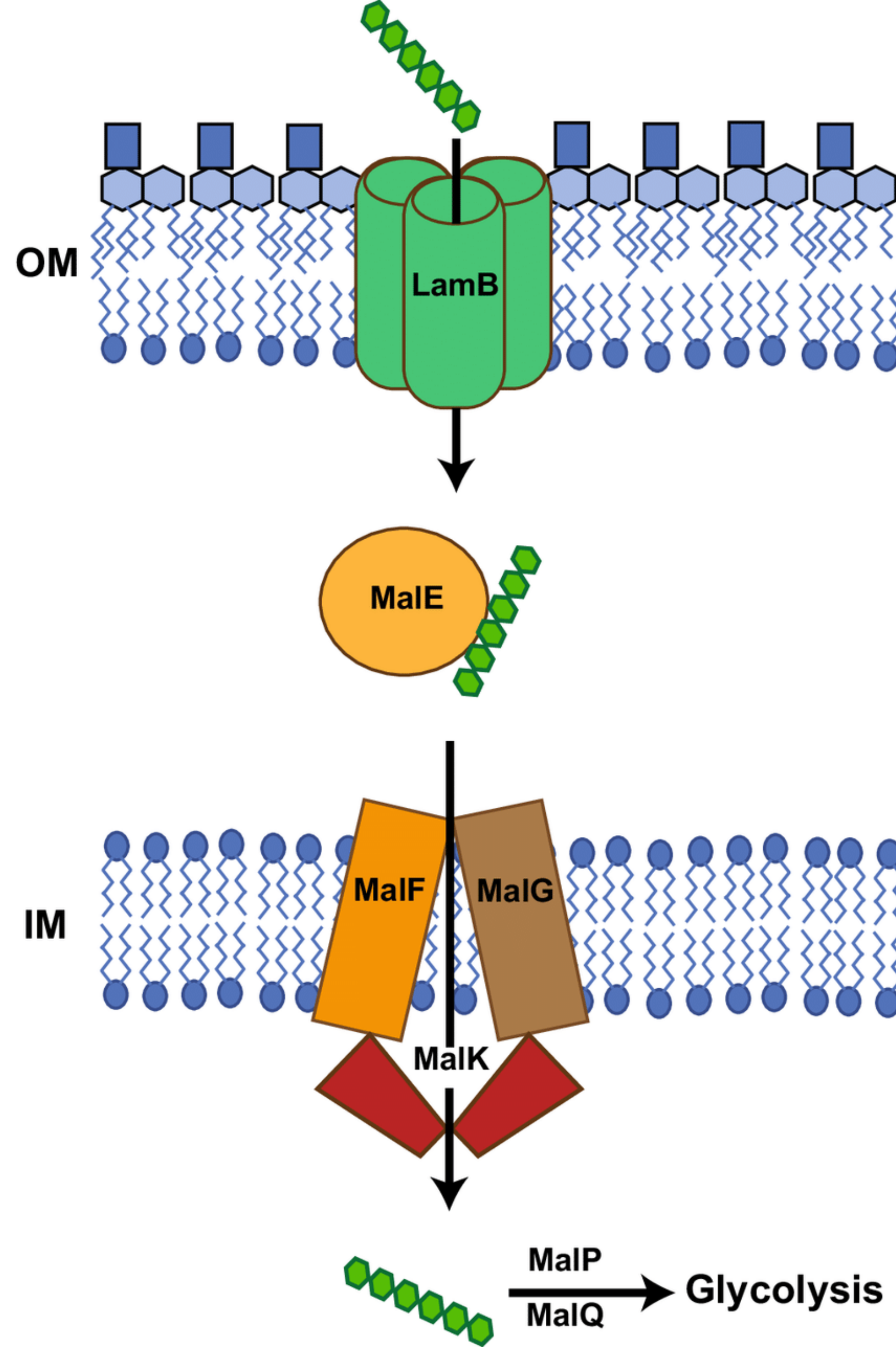
An optimal compromise

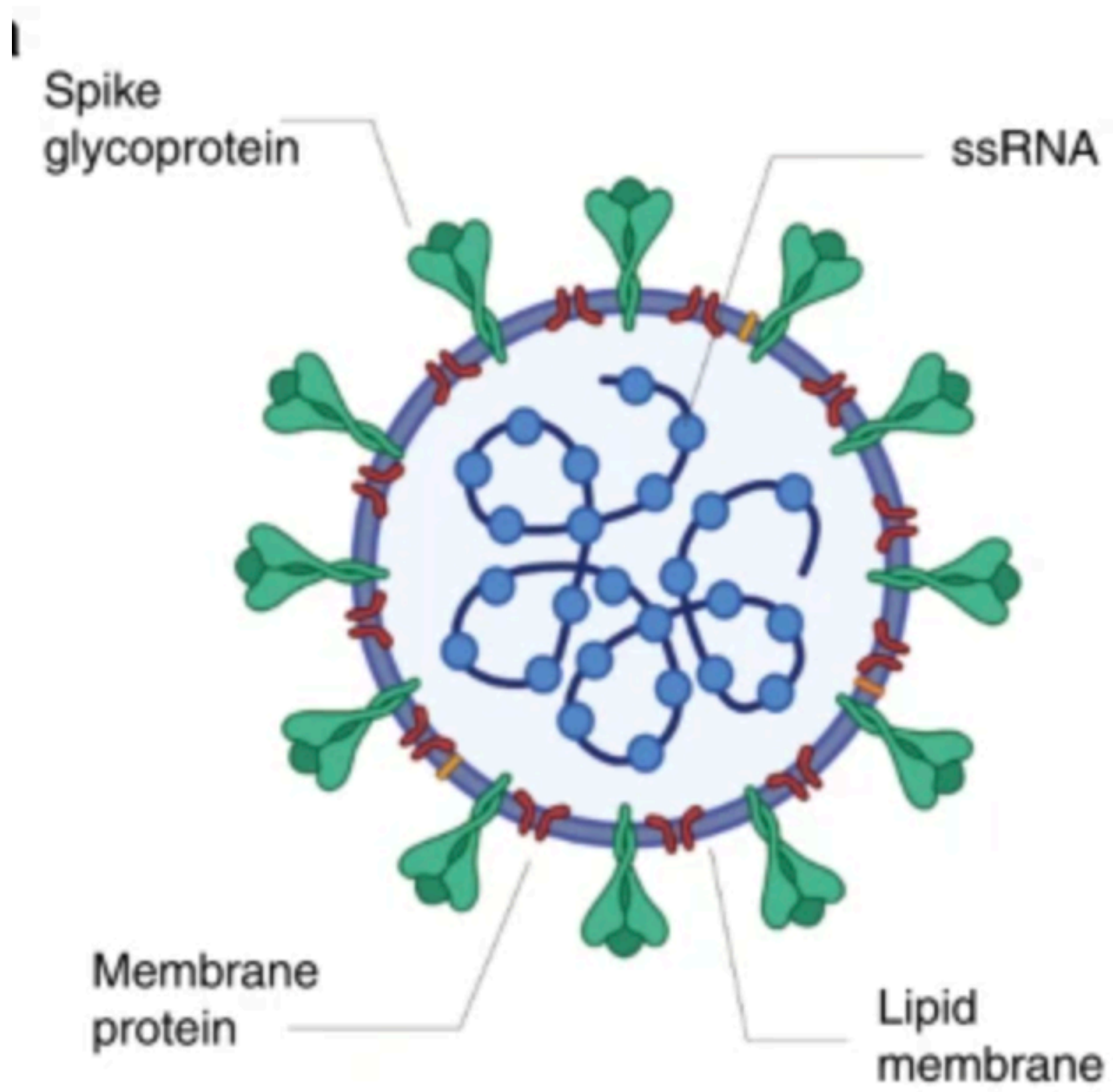
$$\eta_{opt} = \sqrt{\omega \eta_0}$$

$$\beta_{opt} = \beta_0 \frac{\sqrt{\eta_0}}{\sqrt{\omega} + \sqrt{\eta_0}}$$

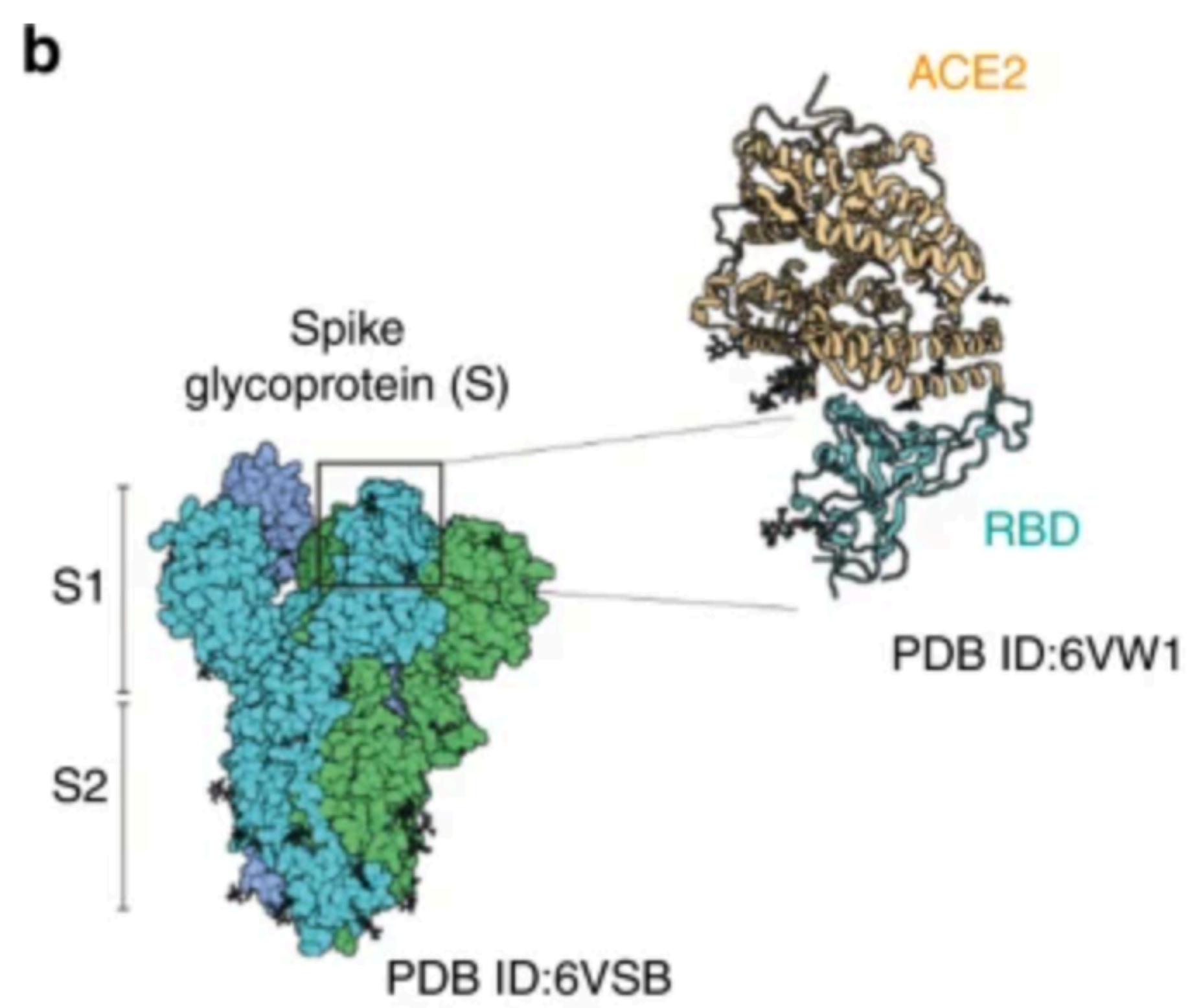
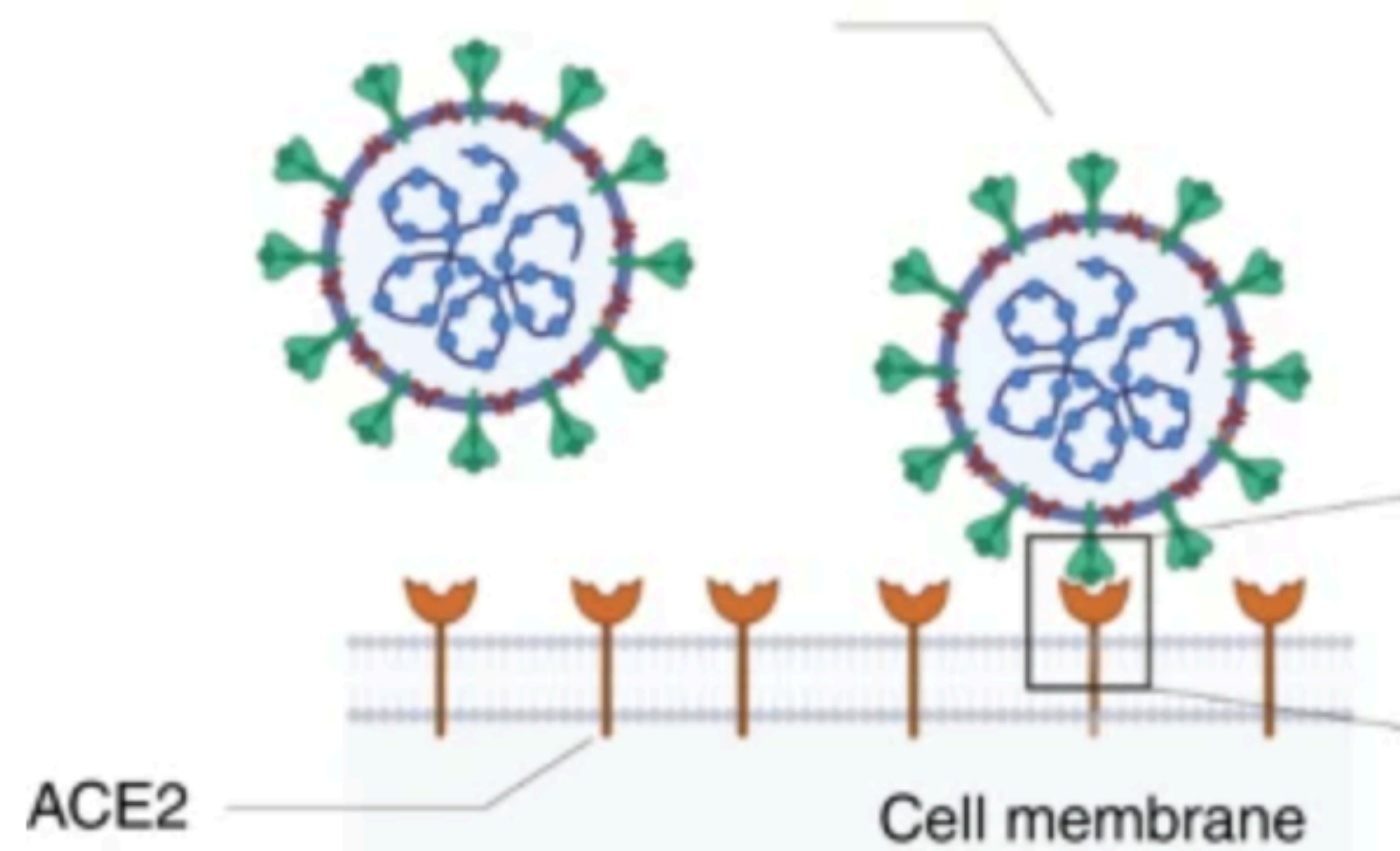
Coevolution







SARS-CoV-2



Coevolution